DES649: Computer-Aided Design and Prototyping

Course Project — Innovative Product Design and Prototype Development

Weightage: 30% of the course

1. Project Overview

The course project requires students to identify a meaningful design opportunity and develop an **innovative product, component, mechanism, or system** through a complete **CAD-to-prototype development cycle**.

Students are expected to move beyond conventional CAD modelling and demonstrate how computer-aided design can be used to **conceive, develop, analyse, refine, and physically realise a product or prototype**.

The project should ideally result in one or more of the following:

- A functional physical prototype;
- A novel product, mechanism, device, or design solution;
- A **patentable innovation**;
- A research contribution suitable for a peer-reviewed publication.

Students are strongly encouraged to pursue **both a patent and a research paper** where the technical novelty and research contribution justify them.

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2. Project Objective

The project aims to enable students to integrate the concepts learned throughout DES649, including:

- Computer-aided geometric modelling;
- Solid and surface modelling;
- Parametric CAD;
- Design for manufacturing and prototyping;
- Rapid prototyping and additive manufacturing;
- Design iteration and optimisation;
- Prototype evaluation and validation;
- Engineering communication and documentation.

The central philosophy of the project is:

Identify a problem → conceptualise an innovative solution → develop the CAD model → prototype → test → iterate → validate → document the innovation.

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3. Expected Project Outcome

Each project should demonstrate a clear progression through the following stages:

Problem Identification
↓
Design Requirements & Specifications
↓
Concept Generation
↓
Concept Selection
↓
CAD Development
↓
Engineering Analysis / Simulation
↓
Prototype Development
↓
Testing & Evaluation
↓
Design Iteration
↓
Final Product / Prototype

Where appropriate:

Novelty Identification → Patent → Research Paper

The project should therefore not be treated simply as a "CAD modelling assignment." The **physical realisation and evaluation of the design are essential components**.

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4. Project Theme

Students may select a problem from areas such as:

- Assistive and medical devices
- Sustainable products
- Manufacturing and automation
- Robotics
- Mechanisms and machines
- Consumer products
- Educational products
- Agricultural technology
- Smart products
- Ergonomic products

- Sports and fitness
- Disaster-response products
- Mobility and transportation
- Household products
- Product accessibility
- Tools and workshop equipment
- Sustainable manufacturing
- Bio-inspired products
- Human–machine interaction

Students are encouraged to identify **real problems faced by users, industries, communities, or organisations**.

Projects that merely reproduce an existing commercial product without a significant innovation will receive limited credit.

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5. CAD Requirements

The project must demonstrate substantive use of CAD.

Students should develop a sufficiently detailed digital representation of the proposed product, including, as appropriate:

- 3D solid models;
- Surface models;
- Parametric models;
- Assemblies;
- Engineering drawings;
- Exploded views;
- Sectional views;
- Mechanism representations;
- Manufacturing details;
- Material and component specifications.

Students should demonstrate that the CAD model was used as an **active design tool**, rather than merely as a means of producing a final illustration.

Where relevant, students should use CAD to investigate:

- Dimensions and tolerances;

- Ergonomics;
- Interference;
- Assembly;
- Motion;
- Strength;
- Manufacturability;
- Weight;
- Material usage;
- Design alternatives.

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6. Prototyping Requirements

Every project must include **at least one physical prototype**.

Depending on the project, students may use:

- FDM/FFF;
- SLA/DLP/MSLA;
- SLS;
- Laser cutting;
- CNC machining;
- Sheet-metal fabrication;
- 3D printing;
- Vacuum forming;
- Casting;
- Hybrid fabrication;
- Conventional workshop processes;
- Other appropriate prototyping technologies.

The prototype should be used to **test and learn about the design**.

Students should document:

1. Prototype version 1;
2. Problems identified;
3. Design modifications;
4. Prototype version 2 or subsequent iterations;

5. Testing results;
6. Final design.

Thus, **iteration is an important component of the project**.

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7. Innovation and Patent Opportunity

A major objective of the project is to encourage students to develop **original and potentially protectable innovations**.

Students should therefore consider, throughout the project:

- What is new about the proposed design?
- What problem does the innovation solve?
- How is it different from existing solutions?
- What technical advantage does it provide?
- Could the mechanism, configuration, process, or product potentially be patented?

Students whose projects demonstrate sufficient novelty will be encouraged to pursue **patent filing through the appropriate institutional process**.

Important

A patent is **not guaranteed merely because a student submits a project**. Patentability will depend on factors such as novelty, inventive step/non-obviousness, usefulness, prior art, and the applicable institutional/legal process.

Students should **not publicly disclose potentially patentable inventions before appropriate consultation**, as premature disclosure may affect intellectual-property protection.

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8. Research Paper Opportunity

Students are also encouraged to convert technically strong projects into **research papers**.

Possible publication venues may include journals such as:

- *Computer-Aided Design*
- *Computer-Aided Geometric Design*
- *Rapid Prototyping Journal*
- Other appropriate journals/conferences in CAD, product development, digital manufacturing, design engineering, additive manufacturing, or related fields.

The expected contribution of a paper could involve, for example:

- A novel CAD methodology;
- A new geometric modelling approach;

- A novel product/mechanism;
- A computational design method;
- A design optimisation methodology;
- A new prototyping methodology;
- Experimental validation of a new design;
- Comparison of alternative CAD/prototyping approaches;
- A new design-manufacturing workflow.

A paper should contain a genuine research contribution and not merely document the construction of a student project.

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9. Instructor Mentorship

Projects aiming for **patent filing and/or journal publication will require close interaction with the course instructor**.

Students should meet the instructor at defined stages to discuss:

- Problem definition;
- Prior-art/literature search;
- Novelty;
- CAD strategy;
- Prototype development;
- Experimental methodology;
- Results;
- Patentability;
- Research-paper structure.

Students are expected to take the initiative in developing the work, while the instructor will provide guidance on the **technical direction, research methodology, IP strategy, and scholarly presentation**.

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10. Suggested Project Milestones

Stage	Deliverable	Indicative Timeline
1	Problem identification & project proposal	Week 1
2	Literature/prior-art review & design requirements	Week 1

Stage	Deliverable	Indicative Timeline
3	Concept generation and selection	Week 2
4	Initial CAD model	Week 2
5	CAD analysis / design refinement	Week 3
6	Prototype 1	Week 3
7	Testing & design iteration	Week 4
8	Final prototype & validation	Week 5
9	Final presentation/demo	Week 6
10	Patent/paper development	During and after project

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11. Assessment Scheme — 30 Marks

Component	Marks
Problem identification, motivation & requirements	3
Innovation, originality & novelty	5
CAD modelling and design development	6
Prototype development & fabrication	5
Testing, validation & design iteration	4
Technical documentation & presentation	3
Patent / research contribution	4
Total	**30**

Patent / Research Contribution — 4 Marks

This component can be awarded based on the level of achievement:

- **Strong research/IP potential identified and systematically developed:** 1–2 marks
- **Patent disclosure/application prepared or submitted:** up to 3 marks
- **High-quality research manuscript prepared/submitted with substantial instructor involvement:** up to 4 marks
- **Exceptional projects may demonstrate both patent and publication potential.**

The assessment should focus on the **quality and substantive contribution of the work**, rather than making publication or patent grant itself the sole determinant of marks.

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12. What Will Make a Project Outstanding?

An outstanding project will demonstrate the following:

A real problem + an original idea + rigorous CAD development + a functional prototype + experimental validation + meaningful iteration + potential for IP/research contribution.

For example:

Problem: Existing wheelchair accessory is difficult to adjust
→ **Innovation:** New quick-adjust mechanism
→ **CAD:** Parametric mechanism and assembly
→ **Analysis:** Motion/interference/strength analysis
→ **Prototype:** 3D-printed functional prototype
→ **Testing:** Adjustment time, load and usability
→ **Iteration:** Improved mechanism
→ **Outcome:** Potential patent + research paper

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13. Academic Integrity and IP

All projects must comply with IIT Kanpur's applicable policies concerning:

- Intellectual property;
- Patent filing;
- Research ethics;
- Authorship;
- Use of external datasets/materials;
- Confidential information.

Students must obtain the instructor's consent before submitting or publicly disclosing a patent or research paper arising from the course project.

Where the instructor has made a substantive contribution to the conception, methodology, technical development, analysis, or preparation of the resulting scholarly work, the instructor should be included as a co-author in accordance with applicable authorship and institutional policies.

Accordingly, **students should not independently submit a patent, journal manuscript, or conference paper arising from this project without prior discussion and consent of the instructor.**

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14. Final Submission

Each team/student should submit:

1. **Project report**

2. **CAD files**
3. **Engineering drawings**, where applicable
4. **Prototype/prototype demonstration**
5. **Testing and validation results**
6. **Design-iteration documentation**
7. **Final presentation**
8. **Patent disclosure/application**, if applicable
9. **Research-paper draft**, if applicable